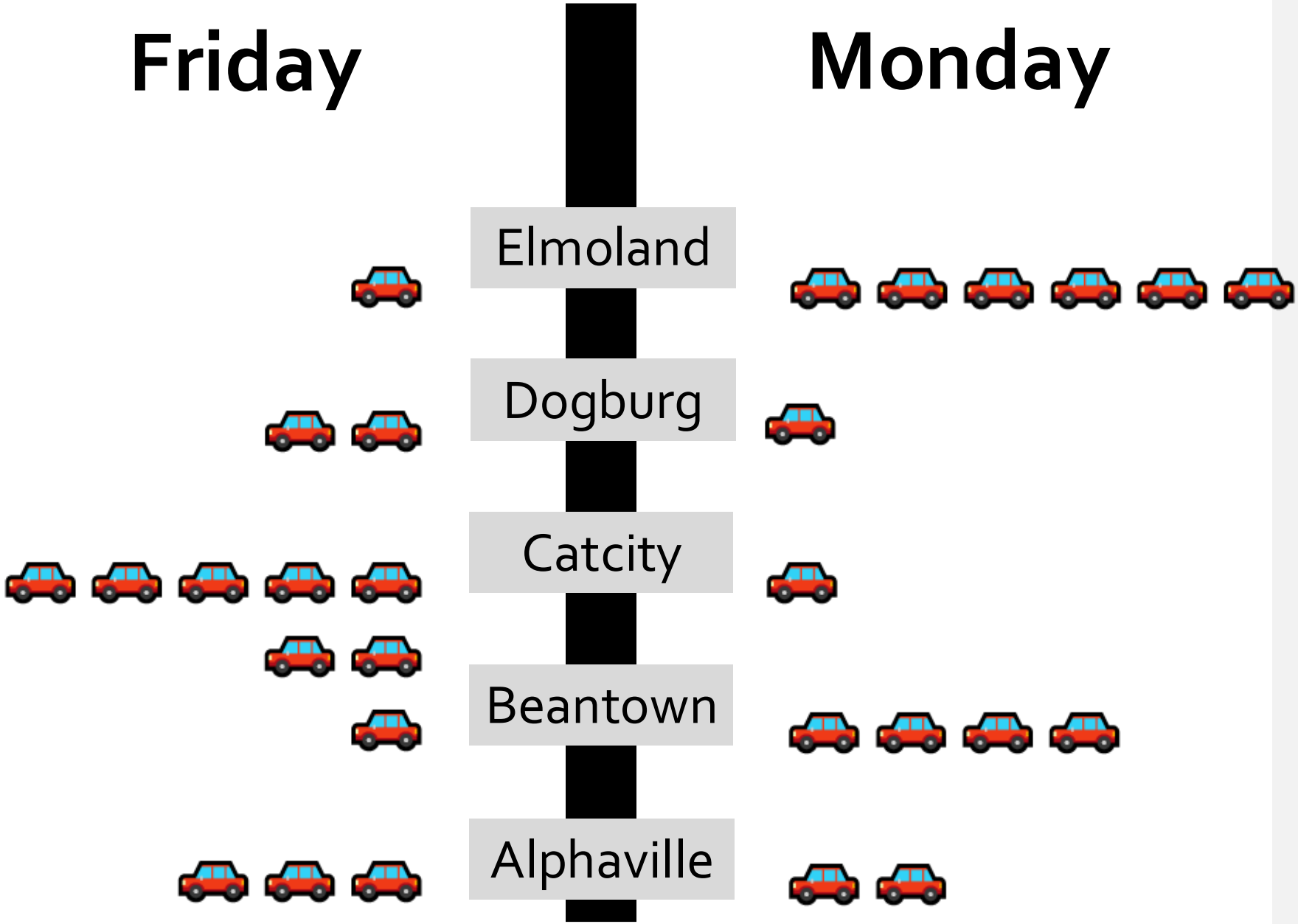
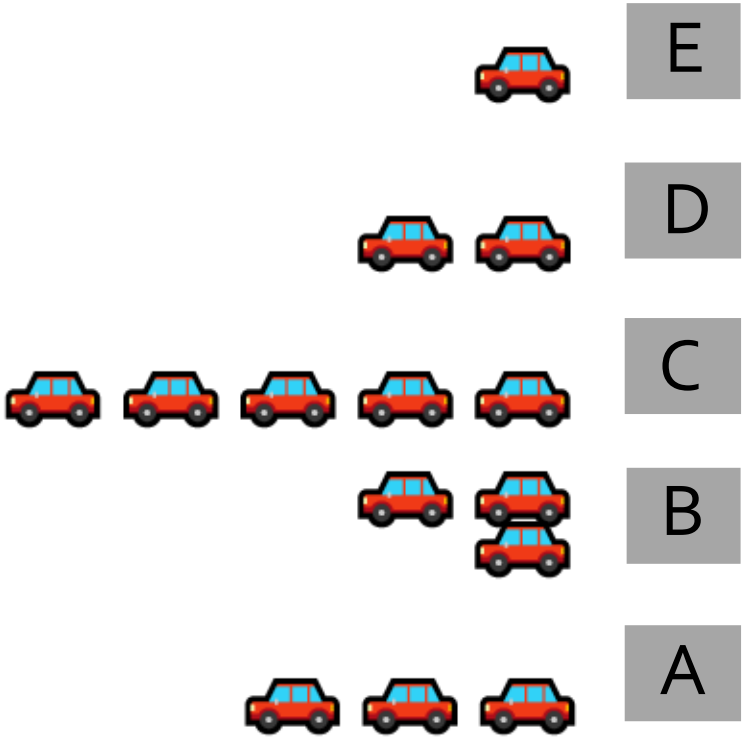
CSI 2101: Discrete Structures

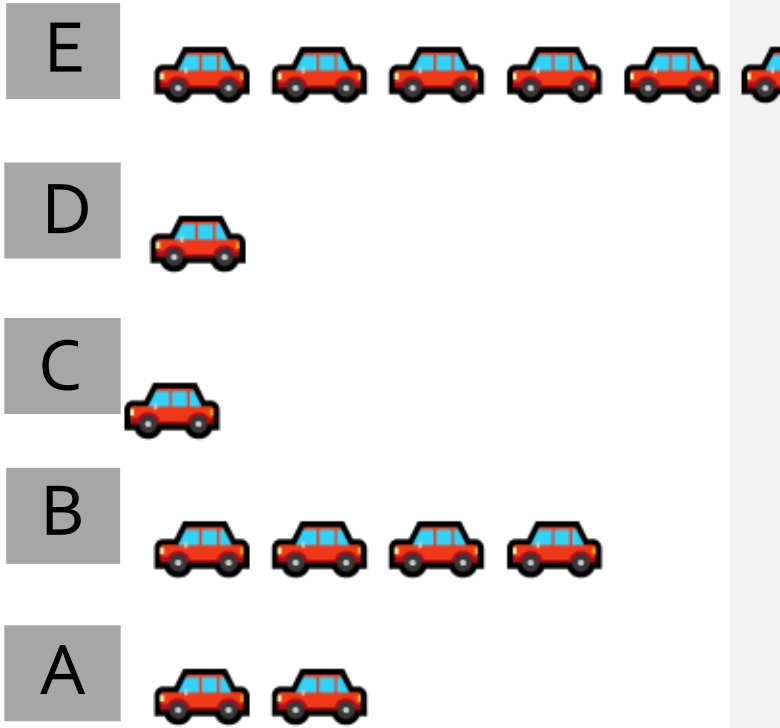
1. LOGIC

Andrej Bogdanov

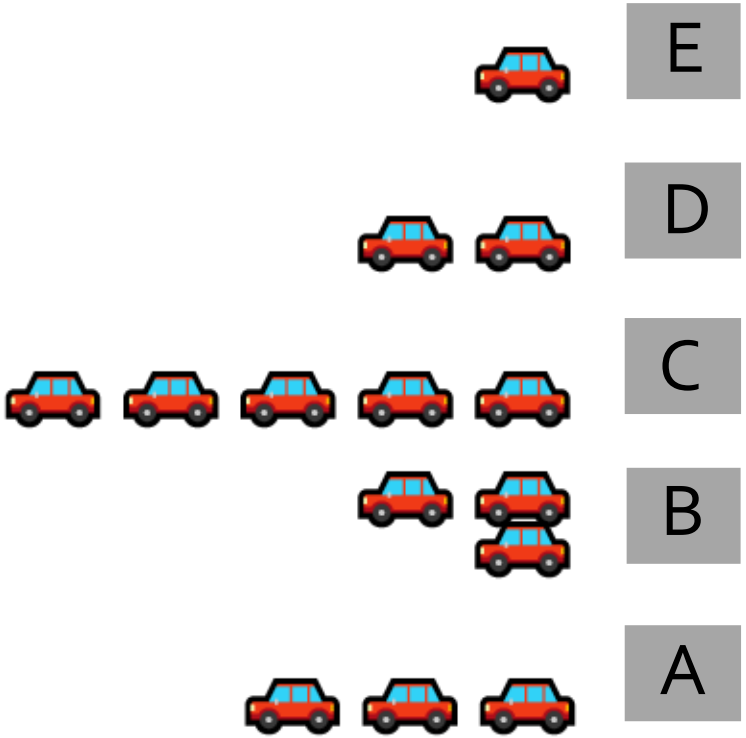

Friday
Monday
Elmoland
Dogburg
Catcity
Beantown
Alphaville

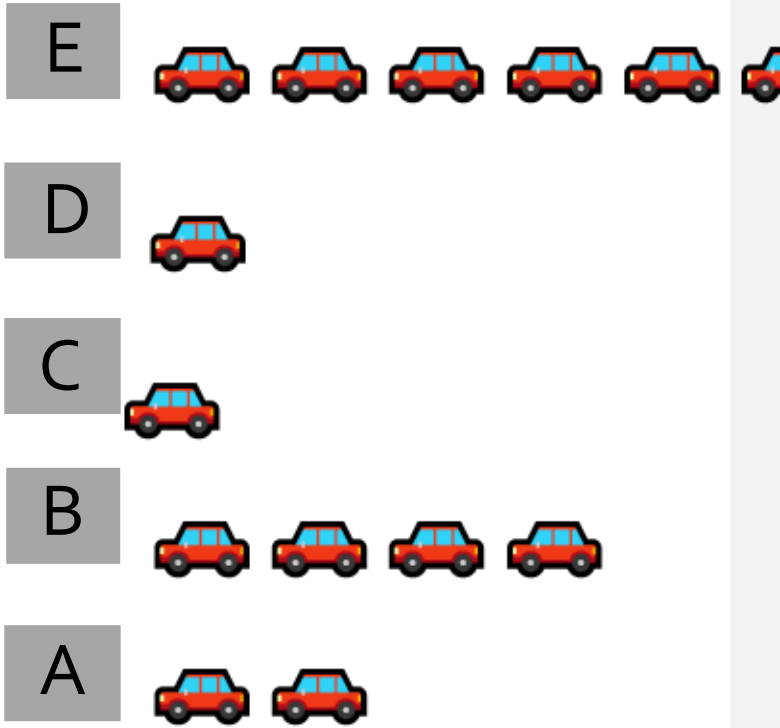
Which car should go where?





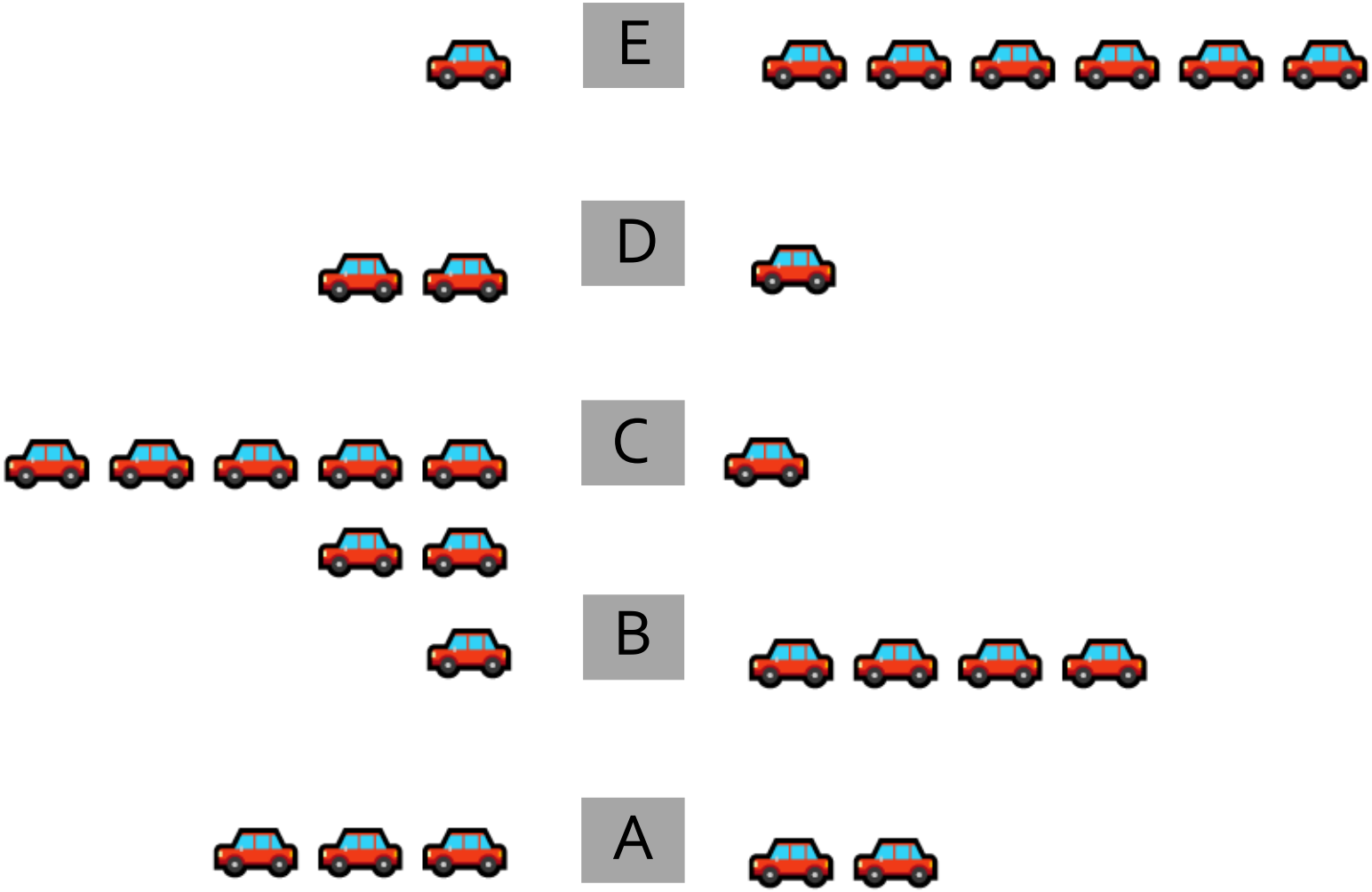
Can you do better?



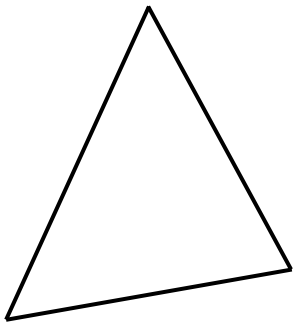


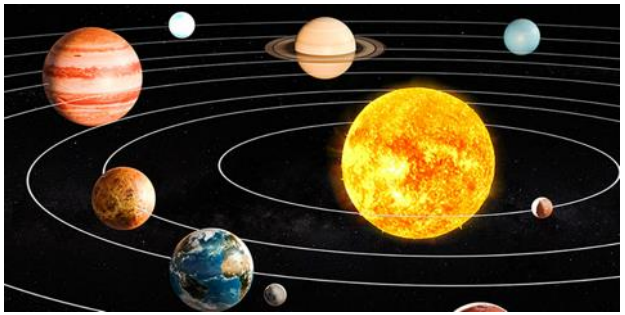
$$2 + 2 = ?$$

Cannot spend < $1200 because



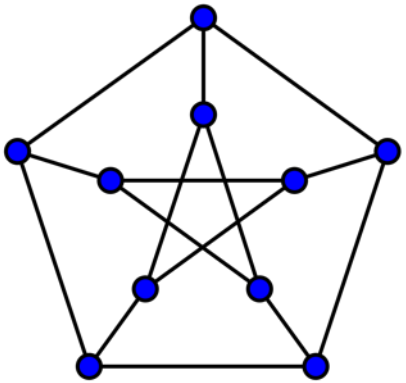
continuous 3.14159265358979323846264 33…

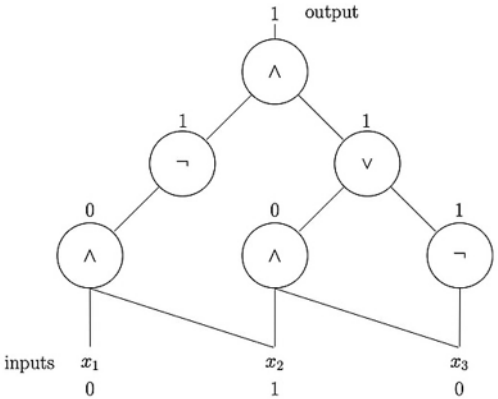




discrete 1001010101001011101 01

42





continuous

$$\int e^{-x} dx$$

$$\nabla \cdot \mathbf{E} = \frac{\rho}{\varepsilon_0}$$

$$\nabla \cdot \mathbf{B} = 0$$

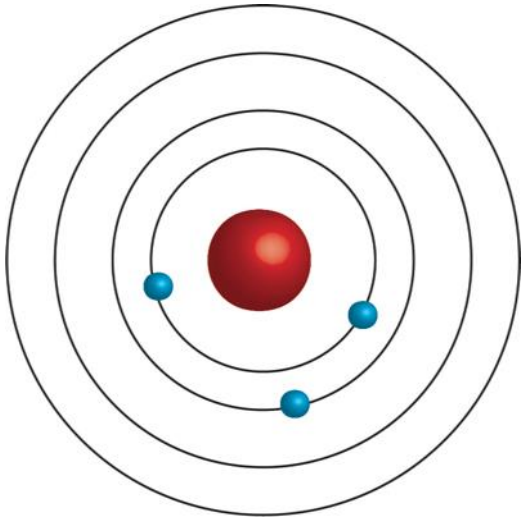
$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$$

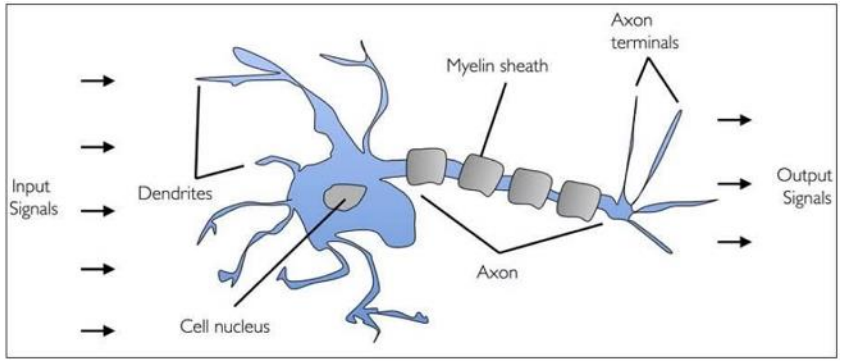
$$\nabla \times \mathbf{B} = \mu_0 \mathbf{j} + \frac{1}{c^2} \frac{\partial \mathbf{E}}{\partial t}$$

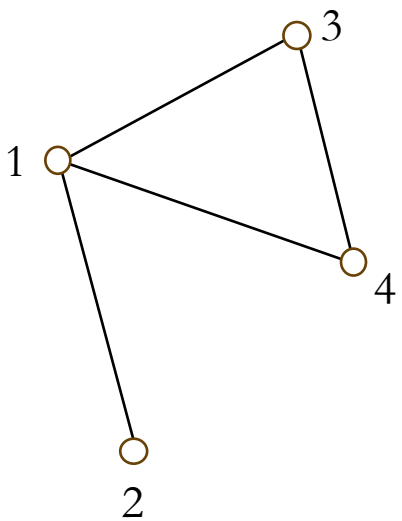
discrete

$$\sum_{n=1}^{\infty} n^{-2}$$

```
s = 0
for i in range(n):
    s += i
return s
```

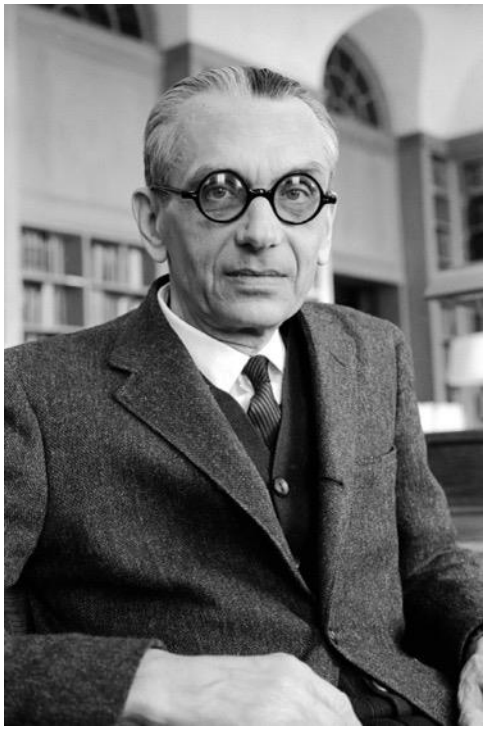


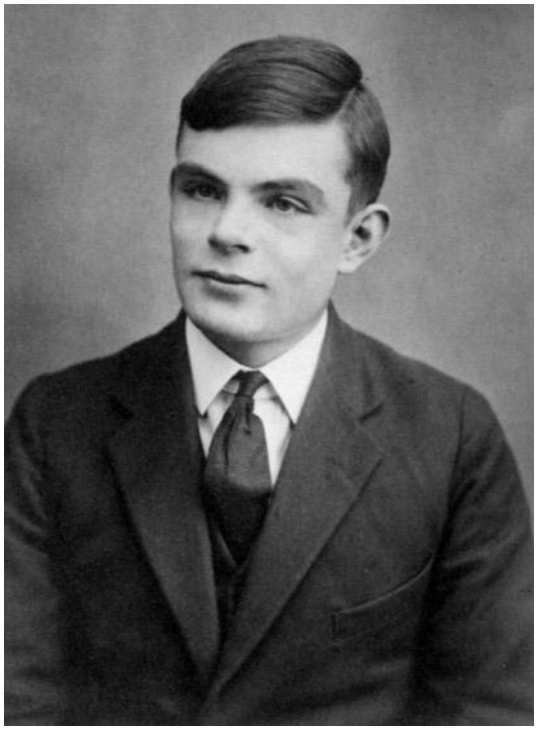

Axon terminals
Myelin sheath
Input Signals
Dendrites
Output Signals
Axon
Cell nucleus



$$\begin{pmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 1 & 0 \end{pmatrix}$$

logic





propositions

$1 + 1 = 3$ F

$1 + 1 = 2$ T

axiomatic method

Start with some propositions that we all agree are true: the axioms

Figure out the rest using the rules of logic

logical operators

NOT (1 + 1 = 3)

(1 + 1 = 2) AND (1 + 1 = 3)

truth tables

NOT

AND

OR

more truth tables

XOR

IFF

Every propositional formula is logically equivalent to one that uses only AND, OR, NOT

Example: P XOR Q

P	Q	P xor Q
T	T	F
T	F	T
F	T	T
F	F	F

predicates

n is even

n is even **IFF** $n + 2$ is even

quantifiers

$\forall$ $\exists$

$\forall n$: n is even

$\forall n$: n is even **IFF** $n + 2$ is even

$\exists n$: n is even

$\exists n$: n is even **IFF** $n + 2$ is even

nested quantifiers

10 is even

$\forall n$: n is even

nested quantifiers

implies

P	Q	$P \rightarrow Q$

$1 + 1 = 2 \longrightarrow 1 + 1 = 3$

$1 + 1 = 3 \longrightarrow 1 + 1 = 2$

$1 + 1 = 3 \longrightarrow 1 + 1 = 4$

from English to propositional logic

(1) Every even number is the sum of two odd numbers.

from English to propositional logic

universe:

relation(s):

1. Alice has friend(s).

2. Alice is a friend of a friend of Bob.

3. Alice and Bob have the same friends.

4. Everyone has a friend.

5. Alice has no friends.

6. Alice has exactly one friend.

7. Alice has more friends than Bob.

from propositional logic to English

universe: classes (CSI 2101, …)
days (Mo, Tu, We, Th, Fr, Sa, Su)

$M(c, d)$: class c meets on day d

1. $\forall c\ \exists d: M(c, d)$

2. $\exists d\, \forall c$: NOT $M(c, d)$

3. NOT $\forall d\, \exists c$: $M(c, d)$

quantifier logic: negation

NOT $\forall x$: $P(x)$

NOT $\exists x$: $P(x)$

4. $\exists d\, \forall c$: $M(c, d)$

THERE IS A DAY WHEN EVERY CLASS MEETS.

F

5. $\forall c\, \exists d$: NOT $M(c, d)$

EVERY CLASS FAILS TO MEET ON AT LEAST ONE DAY.

NOT $\exists c\, \forall d : M(c,d)$

NO CLASS MEETS EVERY DAY.

T

quantifier logic: universal instantiation

$$\exists x\, \forall y\colon P(x, y)$$

quantifier logic: negation

You can please
some of the people all of the time,

and you can please
all the people some of the time,

but you cannot please
all the people all the time

A 3x3 table is filled with different numbers.
Are these predicates equivalent?

(1) x is the largest number in its column and x is the largest number in its row.

(2) x is the largest number in the table.

quantifier logic

Does (1) imply (2)?

(1) $\forall x$: $(P(x)$ **OR** $Q(x))$

(2) $(\forall x$: $P(x))$ **OR** $(\forall x$: $Q(x))$

reasoning about quantifiers

(1) Every even number is the sum of two odd numbers.

(2) For every $n \geq 0$, $n^2 + n + 41$ is a prime.

primes

2	3	5	7	11	13	17	19	23	29
31	37	41	43	47	53	59	61	67	71
73	79	83	89	97	101	103	107	109	113
127	131	137	139	149	151	157	163	167	173

(3) Inscribe a regular n-gon on a circle and draw all diagonals. How many regions are there?

(4a) 29 is the sum of three integer cubes.

(4b) 33 is the sum of three integer cubes.

$$
\begin{aligned}
33 = {} & 8866128975287528^3 \\
& + (-8778405442862239)^3 \\
& + (-2736111468807040)^3
\end{aligned}
$$

A. Booker, *Cracking the problem with 33*, 2022

(5) Every even number greater than 2 is the sum of two primes.